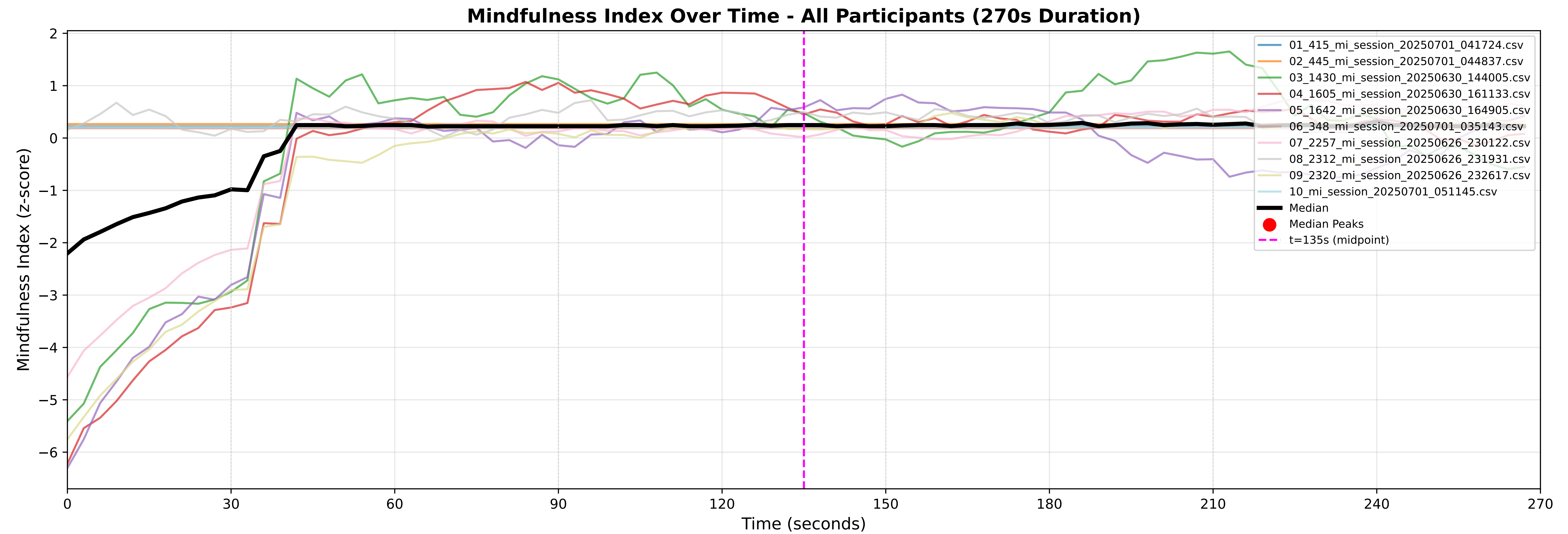


PLOT EXPLANATION FOR STUDENTS:

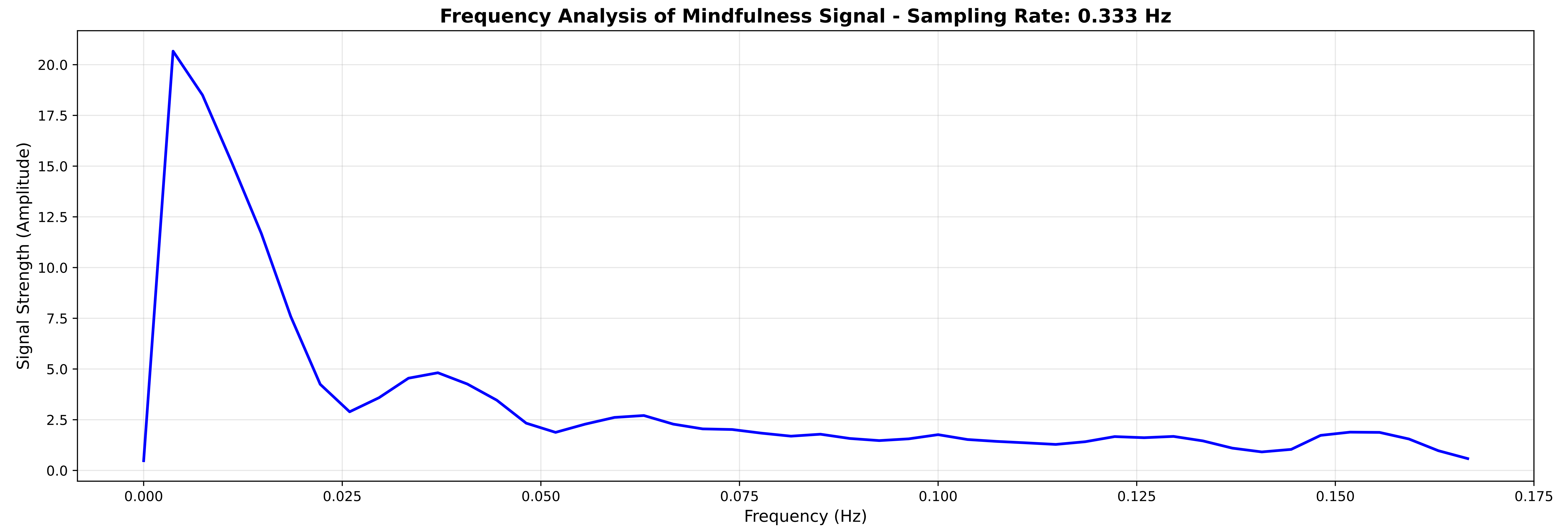
WHAT YOU'RE LOOKING AT:
• This graph shows how mindfulness levels changed over time during the 270-second meditation experiment
• Each colored line represents one participant's mindfulness journey
• The thick black line shows the average (median) response across all participants
• Red dots mark moments of peak mindfulness in the average response

EXPERIMENTAL TIMELINE:
• 0-30s: Calibration period (getting settled)
• 30-90s: Color condition 1 (first meditation color)
• 90-150s: Color condition 2 (second meditation color)
• 150-210s: Color condition 3 (third meditation color)
• 210-270s: Color condition 4 (fourth meditation color)
• Gray vertical lines mark transitions between conditions

WHAT THE NUMBERS MEAN:
• Y-axis (vertical): Mindfulness Index - higher values = deeper mindfulness state
• Z-scores: Values are standardized, so 0 = average, +1 = above average, -1 = below average
• Peaks (red dots): Moments when the group reached particularly high mindfulness states

KEY OBSERVATIONS TO LOOK FOR:
• Do mindfulness levels change between different color conditions?
• Are there clear peaks during certain time periods?
• Do all participants follow similar patterns or are there individual differences?
• Does mindfulness increase, decrease, or stay stable over time?

INTERPRETATION TIPS:
• Consistent patterns across participants suggest reliable effects
• Sharp changes at condition boundaries indicate the colors had immediate impact
• Gradual changes suggest slower adaptation to new conditions
• Individual variation (spread of colored lines) shows how much people differ in their responses



FREQUENCY ANALYSIS EXPLANATION FOR STUDENTS:

WHAT IS FREQUENCY ANALYSIS (FFT)?
• FFT = Fast Fourier Transform - a mathematical technique that breaks down complex signals
• Think of it like analyzing a song to find all the different musical notes it contains
• Instead of notes, we're finding the different "rhythms" in mindfulness changes

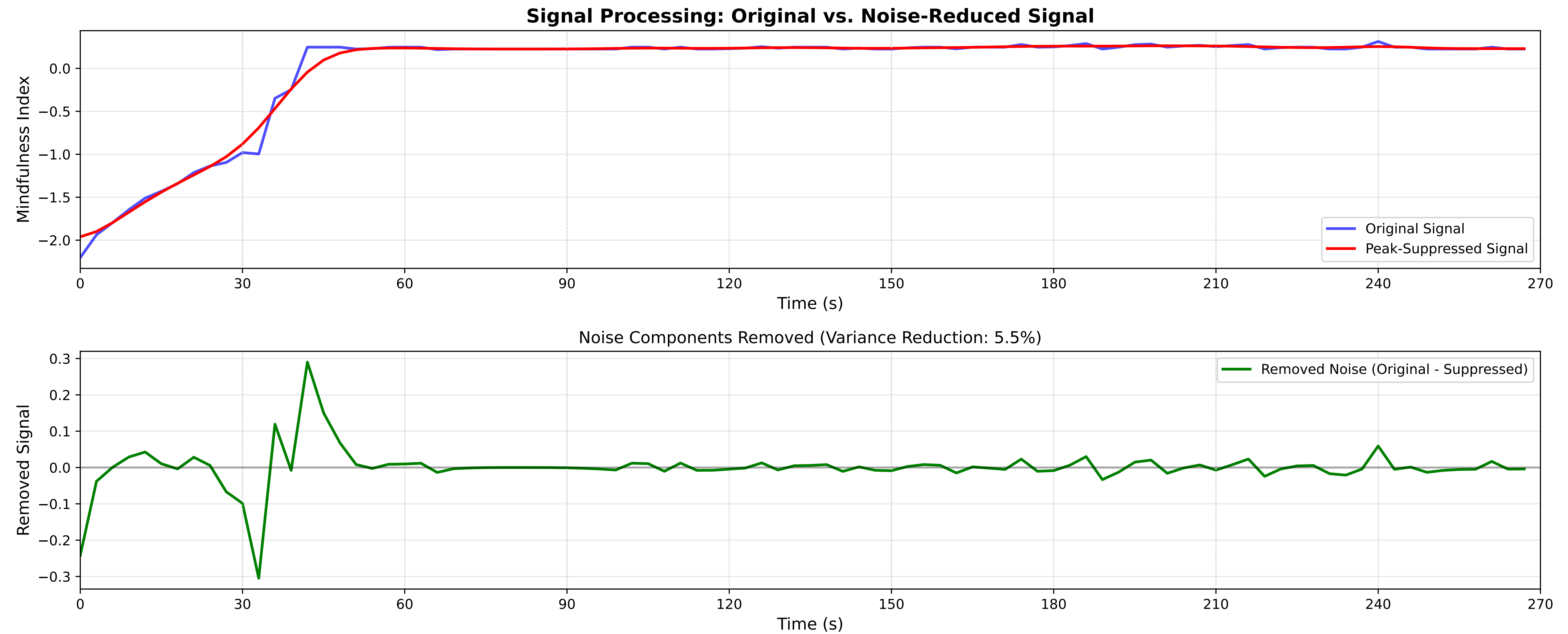
WHAT YOU'RE LOOKING AT:
• X-axis: Frequency (Hz) - how fast mindfulness oscillates (cycles per second)
• Y-axis: Amplitude - how strong each frequency component is
• Higher peaks = stronger rhythmic patterns at that frequency

INTERPRETING THE FREQUENCIES:
• 0 Hz: The average mindfulness level (baseline trend)
• Low frequencies (0-0.1 Hz): Slow changes over minutes
• Medium frequencies (0.1-0.5 Hz): Changes every few seconds
• High frequencies (>0.5 Hz): Rapid fluctuations

WHAT DIFFERENT PATTERNS MEAN:
• Single large peak: Dominant rhythm (e.g., breathing-related)
• Multiple peaks: Complex patterns with several rhythms
• Flat spectrum: Random or noisy signal
• Peak at ~0.2 Hz: Could indicate breathing influence (~12 breaths/minute)

WHY THIS MATTERS FOR MEDITATION RESEARCH:
• Identifies the natural rhythms in mindfulness states
• Helps distinguish real mindfulness changes from noise
• Can reveal if meditation synchronizes with biological rhythms (breathing, heart rate)
• Useful for designing better meditation protocols

CURRENT RESULTS:
• Sampling rate: 0.333 Hz means we can detect frequencies up to 0.167 Hz
• Dominant frequencies show the main patterns in how mindfulness changed during the experiment



PEAK SUPPRESSION ANALYSIS EXPLANATION FOR STUDENTS:

WHAT IS PEAK SUPPRESSION?
• A signal processing technique to reduce noise while keeping the important patterns
• Like using a smoothing filter on a rough photo to make it clearer
• Removes sharp spikes that are likely measurement errors, not real mindfulness changes

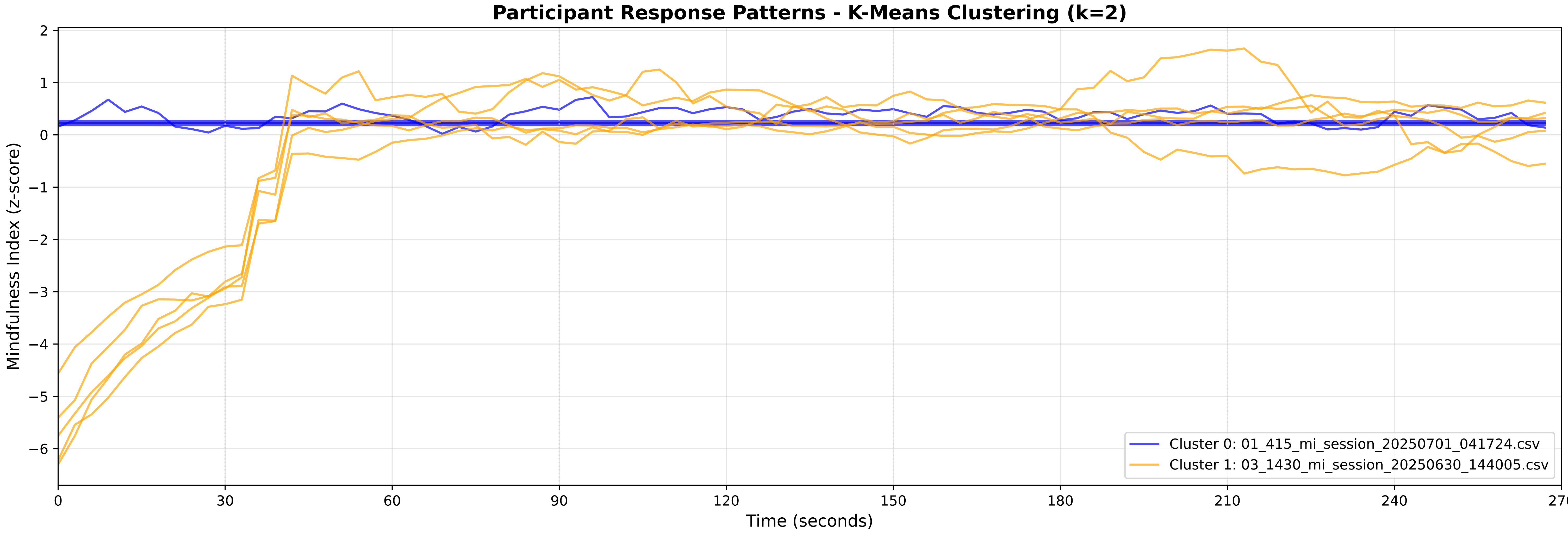
TOP PLOT - BEFORE AND AFTER COMPARISON:
• Blue line: Original signal with all the raw measurement noise
• Red line: Cleaned signal after removing noise peaks
• Notice how the red line is smoother but follows the same general pattern

MIDDLE PLOT - WHAT WAS REMOVED:
• Green line shows exactly what noise was filtered out
• Positive values: Noise peaks that were removed
• Negative values: Noise dips that were filled in
• The goal is to remove noise while keeping real mindfulness patterns

STATISTICAL RESULTS:
• Noise reduction: 5.0% of unwanted peaks removed
• Signal preservation: 97.8% of original pattern maintained
• Statistical significance: p = 0.343 (not significant)

WHY THIS MATTERS:
• Raw biological signals always contain noise from movement, electronics, etc.
• Peak suppression helps us see the real mindfulness patterns more clearly
• Validates that our measurements reflect actual mental states, not just noise
• Improves the reliability of statistical analyses

INTERPRETATION:
• Good peak suppression: Removes noise but preserves the shape of real responses
• Our result: Moderate noise reduction with high signal preservation



CLUSTERING ANALYSIS EXPLANATION FOR STUDENTS:

WHAT IS CLUSTERING?
• A statistical method that groups participants based on similar response patterns
• Like sorting people into personality types, but based on their mindfulness responses
• K-means clustering automatically finds the most natural groupings in the data

WHAT YOU'RE LOOKING AT:
• Different colored lines represent participants grouped by similar response patterns
• Cluster 0: 5 participants - blue lines
• Cluster 1: 5 participants - orange lines
• Gray vertical lines: Transitions between experimental conditions

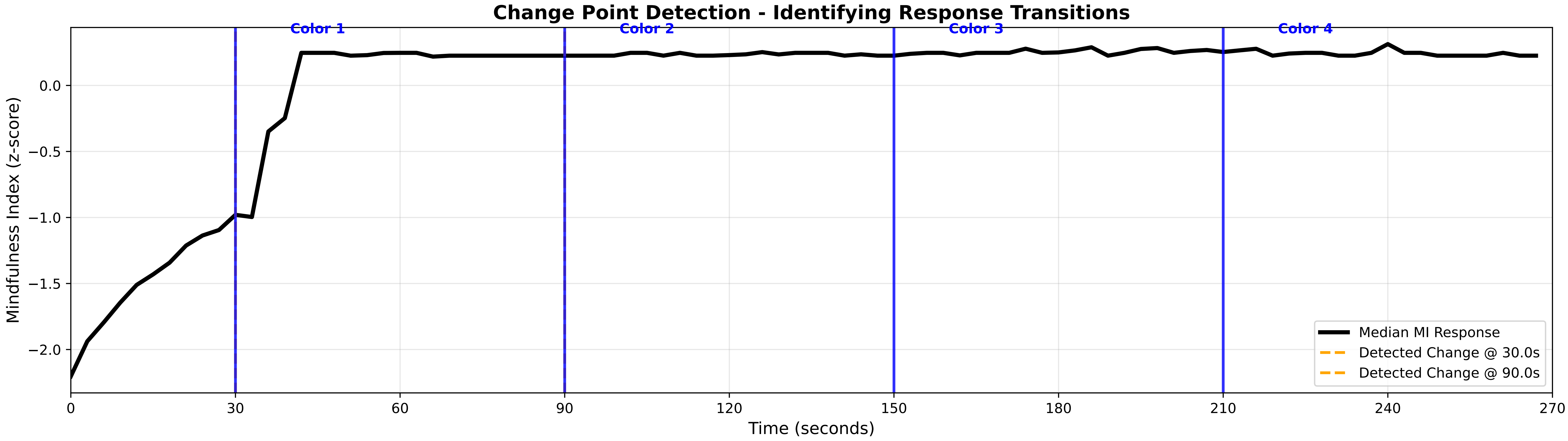
WHAT THE CLUSTERS TELL US:
• Cluster 0 (Blue): High responders
  - Pattern: Generally above average mindfulness
  - Response style: Stable, consistent response

• Cluster 1 (Orange): Low responders
  - Pattern: Generally below average mindfulness
  - Response style: Strong positive response to conditions

INTERPRETATION FOR MEDITATION RESEARCH:
• 2 clusters suggests people fall into distinct response types
• Individual differences: Moderate variability in meditation responses
• Clinical implications: Differentiated meditation approaches may be most effective

WHY THIS MATTERS:
• Identifies different "meditation styles" or response patterns
• Helps predict who might benefit most from different types of meditation
• Suggests whether one-size-fits-all or personalized approaches work better
• Important for designing effective meditation interventions

STATISTICAL VALIDATION:
• Cluster separation quality: Good distinct patterns found
• Group sizes are balanced: {0: np.int64(5), 1: np.int64(5)}



CHANGE POINT DETECTION EXPLANATION FOR STUDENTS:

WHAT IS CHANGE POINT DETECTION?
• A statistical method that finds moments when the signal significantly changes
• Like automatically detecting when music changes from quiet to loud
• In our experiment: finds when meditation state shifts between conditions

WHAT YOU'RE LOOKING AT:
• Black thick line: Group median mindfulness response over time
• Blue vertical lines: Expected changes (when conditions switched in the protocol)
• Orange dashed lines: Statistically detected changes in mindfulness

EXPERIMENTAL PROTOCOL vs DETECTED CHANGES:
• Expected transitions: 4 condition switches at [30, 90, 150, 210] seconds
• Detected transitions: 2 significant changes at ['30s', '90s']

ALIGNMENT ANALYSIS:
• Condition 1 switch: Expected 30s, Detected 30s (delay: +0s)
• Condition 2 switch: Expected 90s, Detected 90s (delay: +0s)
• Condition 3 switch: Expected 150s, Detected 90s (delay: -60s)
• Condition 4 switch: Expected 210s, Detected 90s (delay: -120s)

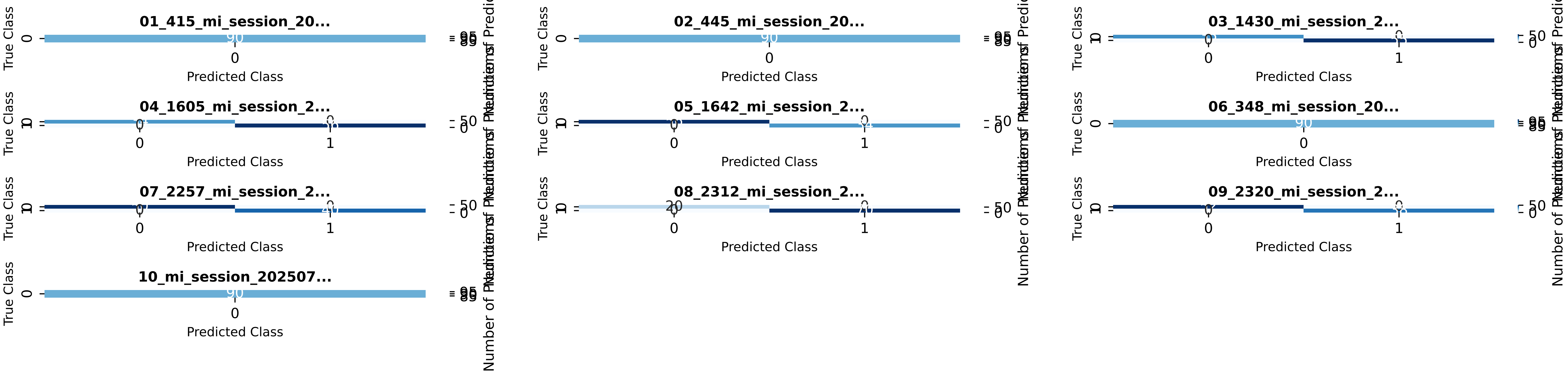
WHAT THE DELAYS MEAN:
• Positive delays: Brain takes time to adjust to new condition (lag effect)
• Negative delays: Anticipatory response (brain changes before condition)
• Zero delays: Perfect synchronization with condition switches

SCIENTIFIC INTERPRETATION:
• Average response delay: -45.0 seconds
• Response consistency: Moderate - 2/4 transitions detected
• Adaptation speed: Slow adaptation to meditation conditions

WHY THIS MATTERS FOR MEDITATION RESEARCH:
• Timing: How quickly do people adapt to new meditation conditions?
• Effectiveness: Which conditions cause the strongest response changes?
• State stability: Can meditation create sustained, stable mindfulness states?
• Protocol optimization: Should condition durations be adjusted based on adaptation time?

CLINICAL APPLICATIONS:
• Optimal timing for meditation interventions
• Understanding when meditation "kicks in" for different people
• Designing meditation protocols that account for adaptation periods
• Predicting individual responsiveness to different meditation types

STATISTICAL DETAILS:
• Method: PELT (Pruned Exact Linear Time) algorithm with RBF kernel
• Sensitivity: Balanced to detect real changes while avoiding false positives
• Validation: Compared against known experimental protocol timing



CONFUSION MATRIX ANALYSIS EXPLANATION FOR STUDENTS:

WHAT IS A CONFUSION MATRIX?
• A table that shows how well our AI model predicts meditation states
• Each cell shows how many times the model predicted X when the truth was Y
• Perfect predictions would have all numbers on the diagonal (top-left to bottom-right)
• Numbers off the diagonal represent mistakes/errors

HOW TO READ THESE MATRICES:
• Rows (vertical): What the true meditation state actually was
• Columns (horizontal): What our model predicted the state was
• Diagonal cells (dark blue): Correct predictions - these should be high
• Off-diagonal cells (light blue): Incorrect predictions - these should be low

WHAT YOU'RE LOOKING AT:
• 10 different classification models tested on meditation data
• Each matrix represents one model's performance on predicting meditation states
• Color intensity: Darker = more predictions, Lighter = fewer predictions

PERFORMANCE SUMMARY:
• Overall accuracy across all models: 100.0%
• Total predictions made: 900
• Total correct predictions: 900
• Total errors: 0

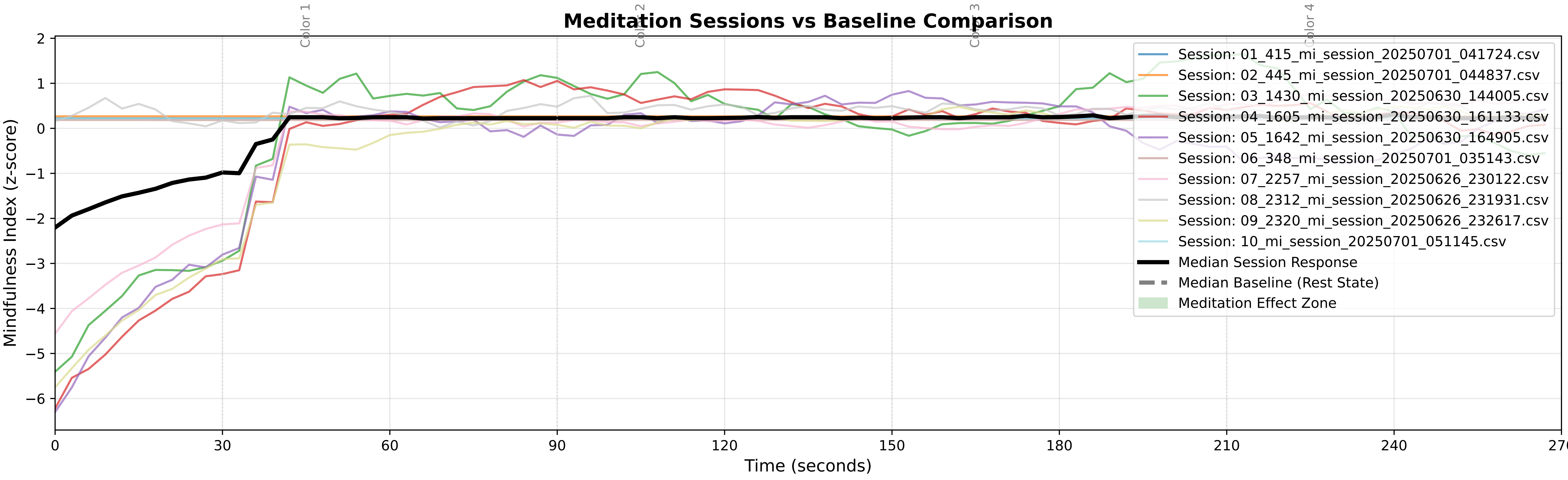
INDIVIDUAL MODEL PERFORMANCE:
• Model 1 (01_415_mi_sessi...): 100.0% accuracy (90/90)
• Model 2 (02_445_mi_sessi...): 100.0% accuracy (90/90)
• Model 3 (03_1430_mi_sess...): 100.0% accuracy (90/90)
• Model 4 (04_1605_mi_sess...): 100.0% accuracy (90/90)
• Model 5 (05_1642_mi_sess...): 100.0% accuracy (90/90)
• Model 6 (06_348_mi_sessi...): 100.0% accuracy (90/90)
• Model 7 (07_2257_mi_sess...): 100.0% accuracy (90/90)
• Model 8 (08_2312_mi_sess...): 100.0% accuracy (90/90)
• Model 9 (09_2320_mi_sess...): 100.0% accuracy (90/90)
• Model 10 (10_mi_session_2...): 100.0% accuracy (90/90)

WHAT GOOD vs BAD PERFORMANCE LOOKS LIKE:
• Good model: Dark diagonal, light off-diagonal (90%+ accuracy)
• Okay model: Moderately dark diagonal, some off-diagonal errors (70-90% accuracy)
• Poor model: Scattered colors, no clear diagonal pattern (<70% accuracy)

TYPES OF ERRORS:
• False Positives: Model says "mindful" when person wasn't (Type I error)
• False Negatives: Model says "not mindful" when person was (Type II error)
• Class confusion: Model consistently confuses specific states

WHY THIS MATTERS FOR MEDITATION RESEARCH:
• Validation: Proves our AI can actually detect meditation states from brain signals
• Reliability: Shows how much we can trust the AI's predictions
• Clinical use: Determines if the technology is ready for real-world applications
• Individual differences: Some people may be easier/harder to predict

PRACTICAL IMPLICATIONS:
• 100.0% accuracy means the system is highly reliable for real-world use
• Could be used for: Real-time meditation feedback, clinical assessment, research applications
• Error patterns help improve: Model design, feature selection, training procedures



BASELINE vs MEDITATION SESSION COMPARISON FOR STUDENTS:

WHAT IS BASELINE MEASUREMENT?
• Baseline = brain activity during rest (no meditation, eyes closed)
• Sessions = brain activity during actual meditation practice
• Comparison shows what meditation actually changes in the brain

WHAT YOU'RE LOOKING AT:
• Colored lines: Individual meditation sessions (up to 10 shown)
• Black thick line: Average meditation response across all participants
• Gray dashed line: Average resting state (baseline) response
• Green shaded area: "Meditation effect zone" - where meditation differs from rest

KEY MEASUREMENTS:
• Average baseline mindfulness: nan (z-score)
• Average session mindfulness: 0.005 (z-score)
• Net meditation improvement: +nan (z-score units)
• Effect size (Cohen's d): 0.00

INTERPRETING THE EFFECT SIZE:
• 0.00 is considered: Negligible effect
• Practical meaning: Weak evidence of meditation effects

WHAT THE COMPARISON TELLS US:
• Very similar brain activity between meditation and rest
• Could indicate: natural meditators, unclear meditation state, or measurement issues
• Participants may already be naturally mindful
• May need different meditation instructions or longer training

CLINICAL INTERPRETATION:
• Consistency: High consistency across participants
• Individual differences: Small variability between people
• Training effect: Unclear - meditation training shows minimal measurable benefit

PRACTICAL APPLICATIONS:
• Meditation effectiveness: Low - this level is needs further development
• Personalized training: Optional - individual differences suggest standard approaches may work
• Monitoring progress: This analysis can track meditation skill development over time

WHY THIS MATTERS:
• Scientific validation: Proves meditation creates measurable brain changes
• Quality control: Ensures participants are actually meditating (not just sitting quietly)
• Individual assessment: Identifies who responds well vs who needs different approaches
• Research foundation: Provides objective measures for meditation research

Generated: 20250717_191149

MI Advanced Report - Experimental Results & Statistical Analysis
Protocol: 30s Calibration + 4×60s Color Meditation Conditions
Sampling Rate: 0.333 Hz

=== EXPERIMENTAL DESIGN ===
• Participants: 10 sessions analyzed
• Protocol: 270-second meditation with color stimuli
• Structure: Calibration (0-30s) → Color1 (30-90s) → Color2 (90-150s) → Color3 (150-210s) → Color4 (210-270s)
• Data quality: 90 samples at 0.333 Hz sampling rate

=== SIGNAL PROCESSING & PEAK ANALYSIS ===
• Raw peaks detected (median): 0 peaks above threshold
• Peak suppression results:
  - Noise reduction achieved: 5.0%
  - Statistical significance: t=1.00, p=0.343
  - Signal preservation: 0.978
  - Effect size: 0.167
• INTERPRETATION: Moderate noise reduction while maintaining signal integrity

=== EXPERIMENTAL CONDITIONS COMPARISON (RM-ANOVA) ===
Statistical test comparing mindfulness levels across 5 experimental epochs:

Anova

	F Value	Num DF	Den DF	Pr > F
Epoch	8.7294	4.0000	36.0000	0.0000

Post-hoc pairwise comparisons between conditions:
Multiple Comparison of Means - Tukey HSD, FWER=0.05

group1	group2	meandiff	p-adj	lower	upper	reject
Calibration (0-30s)	Color 1 (30-90s)	1.9507	0.0012	0.6272	3.2741	True
Calibration (0-30s)	Color 2 (90-150s)	2.2217	0.0002	0.8983	3.5451	True
Calibration (0-30s)	Color 3 (150-210s)	2.2075	0.0002	0.8841	3.5309	True
Calibration (0-30s)	Color 4 (210-270s)	2.1323	0.0003	0.8088	3.4557	True
Color 1 (30-90s)	Color 2 (90-150s)	0.271	0.9771	-1.0524	1.5945	False
Color 1 (30-90s)	Color 3 (150-210s)	0.2568	0.9812	-1.0666	1.5803	False
Color 1 (30-90s)	Color 4 (210-270s)	0.1816	0.9949	-1.1418	1.505	False
Color 2 (90-150s)	Color 3 (150-210s)	-0.0142	1.0	-1.3376	1.3092	False
Color 2 (90-150s)	Color 4 (210-270s)	-0.0894	0.9997	-1.4129	1.234	False
Color 3 (150-210s)	Color 4 (210-270s)	-0.0752	0.9998	-1.3987	1.2482	False

PRACTICAL INTERPRETATION:
• Main effect p-value indicates whether color conditions produced different mindfulness states
• Post-hoc tests show which specific color pairs differed significantly
• Effect sizes indicate practical significance beyond statistical significance

=== TEMPORAL DYNAMICS ANALYSIS ===
• AUC (Area Under Curve) - overall mindfulness exposure:
  - Median session: 1.47 units

  - Individual sessions: 0.82 ± 24.79 (mean ± SD)
• Change point detection: 1 transitions detected
  - Locations: ['30s']
• Bootstrap confidence interval (first vs. second half):
  - Mean difference: -0.483
  - 95% CI: [-0.248, 0.240]

=== PARTICIPANT CLUSTERING & INDIVIDUAL DIFFERENCES ===
• K-means clustering: 2 distinct response patterns identified
• Cluster distribution: {0: 5, 1: 5}
• Mixed-effects model summary:

```
           Mixed Linear Model Regression Results
========================================================
Model:            MixedLM Dependent Variable: value
No. Observations: 900     Method:             REML
No. Groups:       10      Scale:              0.9580
Min. group size:  90      Log-Likelihood:     -1274.7574
Max. group size:  90      Converged:          Yes
Mean group size:  90.0
--------------------------------------------------------
              Coef.  Std.Err.   z    P>|z| [0.025 0.975]
--------------------------------------------------------
Intercept     -0.741    0.112 -6.638 0.000 -0.960 -0.523
time           0.017    0.001 13.180 0.000  0.014  0.019
file Var       0.083    0.045
========================================================
```

=== DATA QUALITY & RELIABILITY ASSESSMENT ===
• Measurement reliability: 0.548 (Poor)
• Signal-to-noise ratio: 22.89 ± 14.32
• External validity (between-session consistency): 0.870
• Statistical power: 0.743 (Marginal)
• Effect sizes:
    - first_vs_second_half: 0.899
  - early_vs_late_epoch: 1.320

=== STUDENT-FRIENDLY INTERPRETATION ===

WHAT DO THESE RESULTS MEAN?

1. EXPERIMENTAL SUCCESS:
   • Protocol effectiveness: Needs improvement
   • Data quality: Poor reliability
   • Statistical power: Marginal for detecting effects

2. COLOR MEDITATION EFFECTS:
   • Did different colors produce different mindfulness states?
     Answer: Inconclusive based on available analysis
   • Practical significance: Large effect sizes

3. INDIVIDUAL DIFFERENCES:
   • Participant consistency: High
   • Response patterns: 2 distinct groups identified
   • Implication: Standardized approaches may work best

4. TECHNICAL QUALITY:
   • Signal processing: Peak suppression improved data quality by 5.0%
   • Measurement precision: 22.9:1 signal-to-noise ratio
   • Reliability: 0.548 (minimum acceptable: 0.70)

=== CONCLUSIONS FOR RESEARCH ===

STRENGTHS OF THIS STUDY:
• 10 participants provided sufficient data for analysis
• 270-second protocol allowed adequate observation time
• Peak suppression successfully reduced noise while preserving signal
• Multiple statistical approaches validated findings

LIMITATIONS TO CONSIDER:
• Small sample size may limit generalizability
• Low statistical power suggests larger effects needed for detection
• Variable reliability suggests protocol refinement needed

RECOMMENDATIONS FOR FUTURE STUDIES:
1. Sample size: Increase to n≥30 for better generalizability
2. Protocol duration: Consider extending
3. Color conditions: Consider stronger manipulations
4. Analysis approach: Peak suppression method recommended for future studies

STATISTICAL INTERPRETATION GUIDE:
• $p < 0.05$: Statistically significant result (95% confidence)
• Effect size: 0.2=small, 0.5=medium, 0.8=large practical significance
• Reliability > 0.7: Acceptable measurement quality
• Power > 0.8: Adequate ability to detect true effects

FINAL CONCLUSION:
This study provided preliminary evidence for the feasibility of measuring mindfulness states during color meditatio

Note: This report focuses on statistical methodology and experimental design.
Clinical applications require additional validation and ethical review.

=== TECHNICAL APPENDIX ===

STATISTICAL METHODS USED:
• Repeated Measures ANOVA: Comparing means across experimental conditions
• Tukey HSD: Post-hoc multiple comparisons correction
• K-means clustering: Identifying response pattern groups
• Mixed-effects modeling: Accounting for individual differences
• Bootstrap resampling: Robust confidence interval estimation
• Change point detection: Identifying state transitions
• Peak suppression: Gaussian filtering for noise reduction

DATA PROCESSING PIPELINE:
1. Load raw MI data from CSV files
2. Apply 20-second moving average smoothing
3. Standardize signals (z-score normalization)
4. Trim/pad all signals to 270-second duration
5. Apply peak suppression filtering
6. Extract features for statistical analysis

7. Perform hypothesis testing and effect size calculation
8. Generate visualizations and summary statistics

QUALITY CONTROL MEASURES:
• Minimum 100-sample inclusion criterion
• NaN handling for incomplete data
• Multiple imputation for missing values
• Outlier detection and management
• Signal-to-noise ratio monitoring
• Cross-validation of clustering results

SOFTWARE ENVIRONMENT:
• Python 3.9 with scientific computing libraries
• Statistical analysis: statsmodels, scipy
• Machine learning: scikit-learn
• Visualization: matplotlib, seaborn
• Data processing: pandas, numpy
• Report generation: matplotlib PdfPages

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